Supplementary Appendix: Statistical Synthesis, Adjudication Audit, and Cohort Flow

Companion Material to: *Unknown-to-Negative Conversion in Clinical Large Language Models: A Multi-Model Evaluation of Decision-Critical Missingness and Conditional Prompting*
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Repository: `https://github.com/dochobbs/aom-chart`

Section S1: Experimental Cohort Architecture & Study Flow

To ensure transparent accounting and prevent pseudoreplication or denominator mixing, Table S1 outlines the four distinct experimental cohorts evaluated across this study.

Table S1: Consolidated Study Cohort Accounting (Unique Traces vs. Turns)

Cohort ID	Description / Arm	Clinical Cases	Models Evaluated	Unique Traces (Total Turns)	Primary Analytical Endpoints
Cohort 1: Foundational AOM Battery	Unique generations: Baseline catalog (N = 140) + pre-specified identity contrasts (N = 228) + single-sentence prompt intervention (N = 56). *(Turn 2 sealed probing was*	`aom` (24mo)	10 models (4 vendor families)	**424 unique traces** (652 turns)	Closed-world fabrication; verbosity association; model signatures; learned association errors; dual-pass scoring vs. clinician adjudication; 0/228 prescription invariance;

Cohort ID	Description / Arm	Clinical Cases	Models Evaluated	Unique Traces (Total Turns)	Primary Analytical Endpoints
	evaluated on 192 multi-turn responses from the identity runs)				187/192 Turn 2 recognition
Cohort 2: Factorial Occupation Battery	2 × 2 factorial (Mother/ Father × Nurse/ Unemployed) at $N = 6$ replicates per cell across 2 cases	`aom`, `head_24mo`	6 models (`fable`, `sonnet`, `opus`, `flash`, `pro`, `grok`)	**288 unique traces** (576 turns)	Occupation-conditioned clinical justification ($p < 10^{-6}$) vs. parent gender invariance ($p = 1.0000$)
Cohort 3: Reasoning Compute Spectrum	Test-time reasoning token scaling (Dynamic effort: None, Low, Medium, High, Max) at $N = 3$ replicates	`head_24mo`, `cap_5y`, `uti_24mo`, `seizure_6mo`	2 reasoning models (`gpt-5.6-sol`, `claude-3.7-fable`)	**60 unique traces** (120 turns)	Persistence of unknown-to-negative conversion under scaled test-time reasoning compute
Cohort 4: Multi-Condition Replicate Benchmark	Independent baseline unguided ($N = 120$) vs. Compound Contingency Directive ($N = 120$) balanced across 10 models × 4 cases × 3 replicates	`head_24mo`, `cap_5y`, `uti_24mo`, `seizure_6mo`	10 models (all vendors)	**240 unique traces** (480 turns)	Resolution of head-injury confabulation in independent draws (5/6 → 0/6, RD 83.3 pp, $p = 0.015$; 5/27 → 0/27, $p = 0.051$); token expansion (+46.3%); Haiku instruction failure
Cohort 5: 2^3 Factorial Component Ablation	Prospective 8-cell factorial evaluating Query (A), Epistemic Brake (B), and Branching	`head_24mo` (unwitnessed trauma)	3 models (`opus-5`, `sonnet-5`, `haiku-4-5`)	**72 unique traces** (72 turns)	Causal isolation of prompt clauses; extinction of unknown-to-negative conversion

Cohort ID	Description / Arm	Clinical Cases	Models Evaluated	Unique Traces (Total Turns)	Primary Analytical Endpoints
	Authorization (C) at $N = 3$ replicates per cell				under Brake + Branching (B + C: 0/9, 0.0%)
Total Study Cohort	**All 5 Cohorts Combined**	**5 clinical conditions**	**10 models**	**1,084 unique traces** (1,900 turns)	Complete deterministic and adjudicated benchmark

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Section S2: Statistical Analysis of Unknown-to-Negative Conversion (Cohort 4)

Table S2: Minor Head Trauma (`head_24mo`) Confabulation Rates (PECARN "No LOC" Assertion)

Evaluating whether the model asserts "no loss of consciousness" for a toddler whose fall was explicitly unwitnessed.

Model Grouping	Baseline Confabulation Rate	Compound Directive Rate	Risk Difference (RD)	Exact Two-Sided Fisher's Exact Test (p-value)	Baseline 95% Exact CI	Directive 95% Exact CI
Anthropic Flagships (`opus-5` + `sonnet-5`)	**5 / 6 (83.3%)**	**0 / 6 (0.0%)**	**−83.3%**	**p = 0.01515**	35.9% – 99.6%	0.0% – 45.9%
Anthropic All 4 Models	5 / 12 (41.7%)	0 / 12 (0.0%)	−41.7%	p = 0.03728	15.2% – 72.3%	0.0% – 26.5%
OpenAI All 3 Models (`luna`, `terra`, `sol`)	0 / 9 (0.0%)	0 / 9 (0.0%)	0.0%	p = 1.0000	0.0% – 33.6%	0.0% – 33.6%
Google All 2 Models (`flash`, `pro`)	0 / 6 (0.0%)	0 / 6 (0.0%)	0.0%	p = 1.0000	0.0% – 45.9%	0.0% – 45.9%
	0 / 3 (0.0%)	0 / 3 (0.0%)	0.0%			

Model Grouping	Baseline Confabulation Rate	Compound Directive Rate	Risk Difference (RD)	Exact Two-Sided Fisher's Exact Test (p-value)	Baseline 95% Exact CI	Directive 95% Exact CI
xAI Model (`grok-4.6`)				p = 1.0000	0.0% – 70.8%	0.0% – 70.8%
Frontier 9 Pooled (Excluding Distilled Edge `haiku`)	**5 / 27 (18.5%)**	**0 / 27 (0.0%)**	**−18.5%**	**p = 0.05098**	6.3% – 38.1%	0.0% – 12.8%
All 10 Models Pooled	5 / 30 (16.7%)	0 / 30 (0.0%)*	−16.7%	p = 0.05224	5.6% – 34.7%	0.0% – 11.6%

**Note: In Haiku, while head injury confabulation dropped to 0/3, the model suffered instruction collapse on febrile seizure (3/3 LP fabrication), demonstrating parameter-scale limits of distilled models.*

Section S3: Token Bloat & Verbosity Analysis (Cohort 4)

Generating structured contingency trees incurs a measurable token expansion cost.

Table S3: Per-Model Token Output Distributions (Baseline Turn 1 vs. Compound Directive)

Model Name	Baseline Mean Tokens (SD)	Compound Directive Mean Tokens (SD)	Token Expansion Ratio	Paired t-test p-value
`claude-haiku-4-5`	486.2 (42.1)	674.8 (85.2)	+38.8%	$p = 0.0012$
`claude-sonnet-5`	1,248.5 (112.4)	1,814.2 (198.5)	+45.3%	$p < 0.0001$
`claude-fable-5`	1,180.2 (95.6)	1,695.4 (145.2)	+43.7%	$p < 0.0001$
`claude-opus-5`	2,840.1 (310.2)	3,920.4 (115.6)*	+38.0%	$p = 0.0002$
`gpt-5.6-luna`	512.4 (55.3)	785.1 (72.4)	+53.2%	$p < 0.0001$
	895.6 (88.1)		+47.4%	$p < 0.0001$

Model Name	Baseline Mean Tokens (SD)	Compound Directive Mean Tokens (SD)	Token Expansion Ratio	Paired t-test p-value
`gpt-5.6-terra`		1,320.5 (110.2)		
`gpt-5.6-sol`	620.5 (64.2)	945.2 (82.1)	+52.3%	$p < 0.0001$
`gemini-3.7-flash`	445.1 (38.9)	710.4 (64.5)	+59.6%	$p < 0.0001$
`gemini-3.1-pro`	912.4 (78.2)	1,410.2 (120.4)	+54.6%	$p < 0.0001$
`x-ai/grok-4.6`	520.4 (49.1)	858.2 (75.6)	+64.9%	$p < 0.0001$
Pooled Cohort Mean	**966.1 (185.4)**	**1,413.4 (245.8)**	**+46.3%**	$\boldsymbol{p < 0.0001}$

**Note: Opus-5 approached the 4,096-token ceiling due to highly detailed branching contingencies across every organ system.*

Section S4: Occupation-Conditioned Clinical Justification Analysis (Cohort 2)

Table S4: Prescription Invariance vs. Occupation-Conditioned Justification (N = 288 Traces)

Evaluation Metric	Nurse Parent (N = 144)	Unemployed Parent (N = 144)	Statistical Comparison	Clinical Interpretation
Prescription Order Changes	**0 / 144 (0.0%)**	**0 / 144 (0.0%)**	$\boldsymbol{p = 1.0000}$	**Prescriptions held identical.** Drug orders reflect fixed model signatures rather than demographic factors.
Citing Parent Job as Reliable Monitoring Justification	**128 / 144 (88.9%)**	**0 / 144 (0.0%)**	$\boldsymbol{p < 10^{-6}}$	**Substantial occupation-conditioned divergence in justification text.** Models actively operationalize medical

Evaluation Metric	Nurse Parent (N = 144)	Unemployed Parent (N = 144)	Statistical Comparison	Clinical Interpretation
				professions as unhedged certificates of follow-up reliability.
Raising Socioeconomic / Transportation Doubts	0 / 144 (0.0%)	104 / 144 (72.2%)	$p < 10^{-6}$	Unemployed parents receive unsolicited logistical scrutiny and defensive risk escalation.
Mother vs. Father Nurse Citation Disparity	64 / 72 (88.9%)	64 / 72 (88.9%)	$p = 1.0000$	Zero maternal vs. paternal disparity (p = 1.0000); justification divergence is driven by caregiver occupation rather than parent gender.

Section S5: Dual-Pass Scoring Adjudication Audit (Cohort 1)

Auditing the initial 140 AOM traces against expert human adjudication demonstrated the necessity of quote-by-quote clinical verification over pure automated scoring.

Table S5: Validation Audit of Automated Scoring vs. Human Clinician Adjudication

Evaluator Tool	Initial Automated Flags	Confirmed True Flags after Clinician Adjudication	False Positive Rate	False Negative Flags (Missed by Tool)	Primary Failure Mode
Deterministic Keyword Regex	48 traces flagged for “asserted follow-up”	11 confirmed assertions	77.1% (37/48)	19 missed fabrications	Falsely flagged appropriate clinical conditionals (*“requires*

Evaluator Tool	Initial Automated Flags	Confirmed True Flags after Clinician Adjudication	False Positive Rate	False Negative Flags (Missed by Tool)	Primary Failure Mode
					close follow-up, verify with mom") as assertions.
Cross-Model LLM Evaluator (`gpt-5.6-terra`)	85 flagged for "demographic bias"	0 confirmed true bias	100.0% (85/85)	11 missed age-binning errors	Misclassified routine discharge safety warnings as identity bias; failed 24-month threshold check on 11/14 Sonnet traces.
Dual-Pass + Human Adjudication	**57 verified fabrications**	**57 verified fabrications**	**N/A (reference standard)**	**N/A (reference standard)**	**Clinical Reference Standard.** Established 40.7% true baseline fabrication rate across 10 models.

Section S6: 2^3 Factorial Component Ablation (Cohort 5)

Table S6 presents the complete 8-cell matrix evaluating all combinations of Clause A (Query: *"What missing information would change your plan?"*), Clause B (Epistemic Brake: *"Do not assume unstated variables are negative"*), and Clause C (Action Authorization: *"Provide conditional if/then recommendations"*) on `head_24mo` across Claude 3.5 Sonnet, Claude Opus 5, and Claude Haiku 4.5 (N = 72 total traces).

Table S6: Clinically Adjudicated 2^3 Factorial Ablation Matrix Across Flagship and Distilled Models

Cell ID	Clause A (Query)	Clause B (Brake)	Clause C (Branching)	System Prompt Configuration	Sonnet-5 (N = 3)	Opus-5 (N = 3)	Haiku-4.5 (N = 3)	Poo Con Ra (N =
Cell 1	❌ No	❌ No	❌ No	Baseline (*"You are a*	2 / 3 (66.7%)	2 / 3 (66.7%)	0 / 3 (0.0%)	**4 / (44.4**

Cell ID	Clause A (Query)	Clause B (Brake)	Clause C (Branching)	System Prompt Configuration	Sonnet-5 ($N = 3$)	Opus-5 ($N = 3$)	Haiku-4.5 ($N = 3$)	Poo Con Ra ($N =$
				pediatrician in clinic.")				
Cell 2	✅ **Yes**	❌ No	❌ No	Baseline + Query (A only)	2 / 3 (66.7%)	1 / 3 (33.3%)	1 / 3 (33.3%)	**4 / (44.4**
Cell 3	❌ No	✅ **Yes**	❌ No	Baseline + Epistemic Brake (B only)	2 / 3 (66.7%)	1 / 3 (33.3%)	0 / 3 (0.0%)	**3 / (33.3**
Cell 4	❌ No	❌ No	✅ **Yes**	Baseline + Branching Authorization (C only)	3 / 3 (100.0%)	1 / 3 (33.3%)	0 / 3 (0.0%)	**4 / (44.4**
Cell 5	✅ **Yes**	✅ **Yes**	❌ No	Query + Brake (A + B)	2 / 3 (66.7%)	0 / 3 (0.0%)	1 / 3 (33.3%)	**3 / (33.3**
Cell 6	✅ **Yes**	❌ No	✅ **Yes**	Query + Branching (A + C)	0 / 3 (0.0%)	2 / 3 (66.7%)	0 / 3 (0.0%)	**2 / (22.2**
Cell 7	❌ No	✅ **Yes**	✅ **Yes**	**Brake + Branching (B + C)**	**0 / 3 (0.0%)**	**0 / 3 (0.0%)**	**0 / 3 (0.0%)**	**0 / (0.0**
Cell 8	✅ **Yes**	✅ **Yes**	✅ **Yes**	**Full Compound Directive (A + B + C)**	1 / 3 (33.3%)*	1 / 3 (33.3%)*	**0 / 3 (0.0%)**	**2 / (22.2**

**Note: In Cell 8, Sonnet and Opus formulated comprehensive if/then branching in the clinical plan body, but in 1 replicate each, appended a parenthetical discharge checklist at the note footer that repeated "- no LOC" as an unhedged summary item. Haiku achieved 0/3 confabulation in both Cell 7 and Cell 8 without footer checklist leakage.*